

# TRAFFIC COUNTING AND VEHICLE CATEGORIZATION USING STATIC CAMERA

## Introduction

Rapid urbanization has significantly increased traffic congestion in cities, leading to delays, fuel wastage, and environmental pollution. Traditional traffic monitoring systems often rely on manual surveys or static sensors, which are limited in scalability and flexibility. This project proposes an intelligent traffic congestion analysis system that combines computer vision, real-time visualization, and geospatial mapping to analyze and represent traffic conditions dynamically.

The system uses deep learning-based object detection to extract traffic information from real-world video footage and integrates the processed data into a Geographic Information System (GIS) for spatial and temporal visualization of congestion levels.

## Core Solution Concept

The core solution of this project is to analyze and visualize traffic congestion by integrating deep learning-based object detection, a real-time user interface, and GIS-based spatial mapping.

Traffic video footage is processed using the YOLOv8 object detection model to identify and classify vehicles and pedestrians. A dedicated user interface displays live traffic statistics such as per-second vehicle counts, further classified into cars, motorcycle, trucks, buses, people and traffic state. The processed traffic data is stored in a structured format as CSV file and imported into a GIS environment, where congestion levels are visualized on selected road segments using temporal controls.

This approach enables both real-time traffic monitoring and post-analysis visualization of congestion trends.

## Unique Value Proposition

The proposed system provides an end-to-end traffic analysis pipeline, from raw video capture to spatial congestion visualization.

Unlike conventional traffic analysis systems that focus only on numerical outputs, this solution:

- Uses **deep learning (YOLOv8)** for accurate object detection
- Provides a **real-time dashboard** for immediate interpretation
- Integrates traffic data into **GIS for spatial and temporal analysis**

The combination of real-time monitoring and GIS-based visualization makes the system suitable for operational traffic management as well as long-term urban planning.

## Key Technologies and Data Used

### Technologies

- **YOLOv8** – Real-time object detection and classification
- **Python** – Detection pipeline, data processing, and UI development
- **Tkinter-based User Interface** – Live traffic statistics visualization
- **QGIS** – Spatial and temporal traffic visualization by polygon mapping
- **OpenStreetMap (OSM)** – Base road network data
- **CSV / Excel** – Storage of processed traffic data

## Data Sources

- **Custom-recorded traffic videos** captured by the team at real locations such as:
  - HMT Road
  - NAD Road
 These videos were recorded using fixed camera positions and used as primary input data for the system.

## Traffic Parameters Extracted

- Number of vehicles per second
- Vehicle class counts (cars, buses, motorcycles)
- Pedestrian count
- Average vehicles per second
- Traffic congestion state (Low / Medium / High)
- Traffic congestion value assigned (1, 2, 3)
- Time stamp for temporal analysis

## Methodology (Step-by-Step)

### Step 1: Traffic Video Collection

Traffic videos were recorded at selected road locations (HMT Road, NAD Road, etc.) using fixed camera setups. The camera angle was chosen to cover a specific road segment or junction clearly.



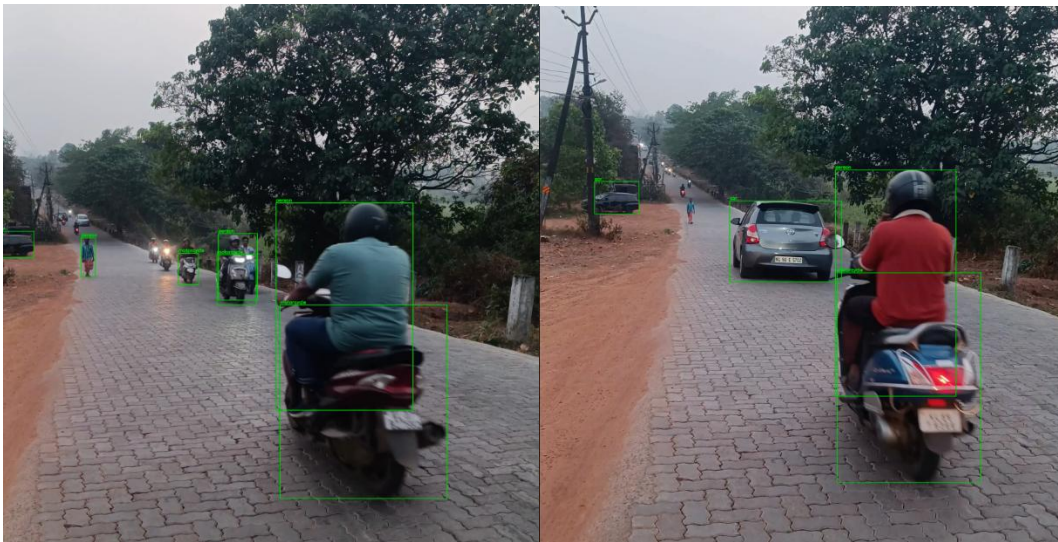
## Step 2: Object Detection Using YOLOv8

The recorded videos were processed frame by frame using the YOLOv8 model. The model detects and classifies objects such as:

- Cars
- Buses
- Motorcycles
- Pedestrians

For each frame:

- Detected objects are classified by type
- Vehicle and pedestrian counts are extracted



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## Step 3: Per-Second Traffic Aggregation

Frame-level detections are aggregated over one-second intervals to compute:

- Average vehicles per second
- Per-class averages
- Total traffic flow rate

Based on predefined thresholds, traffic congestion states are assigned:

- **Low (1)** – Low traffic density
- **Medium (2)** – Moderate traffic density
- **High (3)** – High traffic density

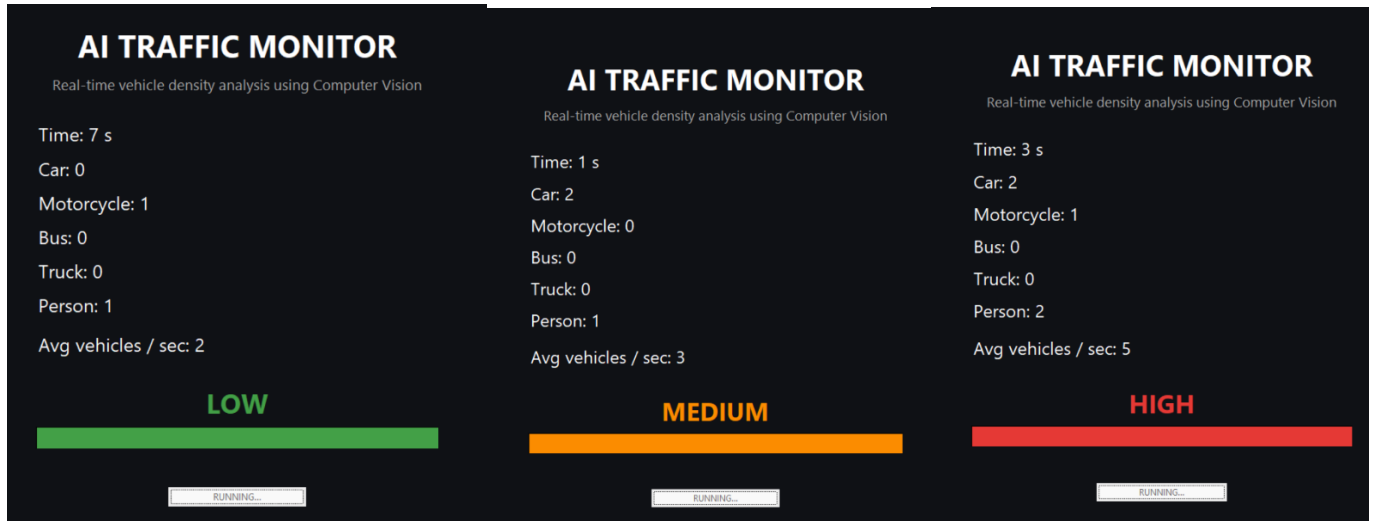
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## Step 4: Real-Time User Interface

A Python-based user interface displays:

- Live time counter
- Vehicle counts per class
- Average vehicles per second
- Current traffic state (Low / Medium / High)
- Visual color indicator (Green / Orange / Red)

This UI allows real-time monitoring and verification of detection accuracy.



## Step 5: Data Storage

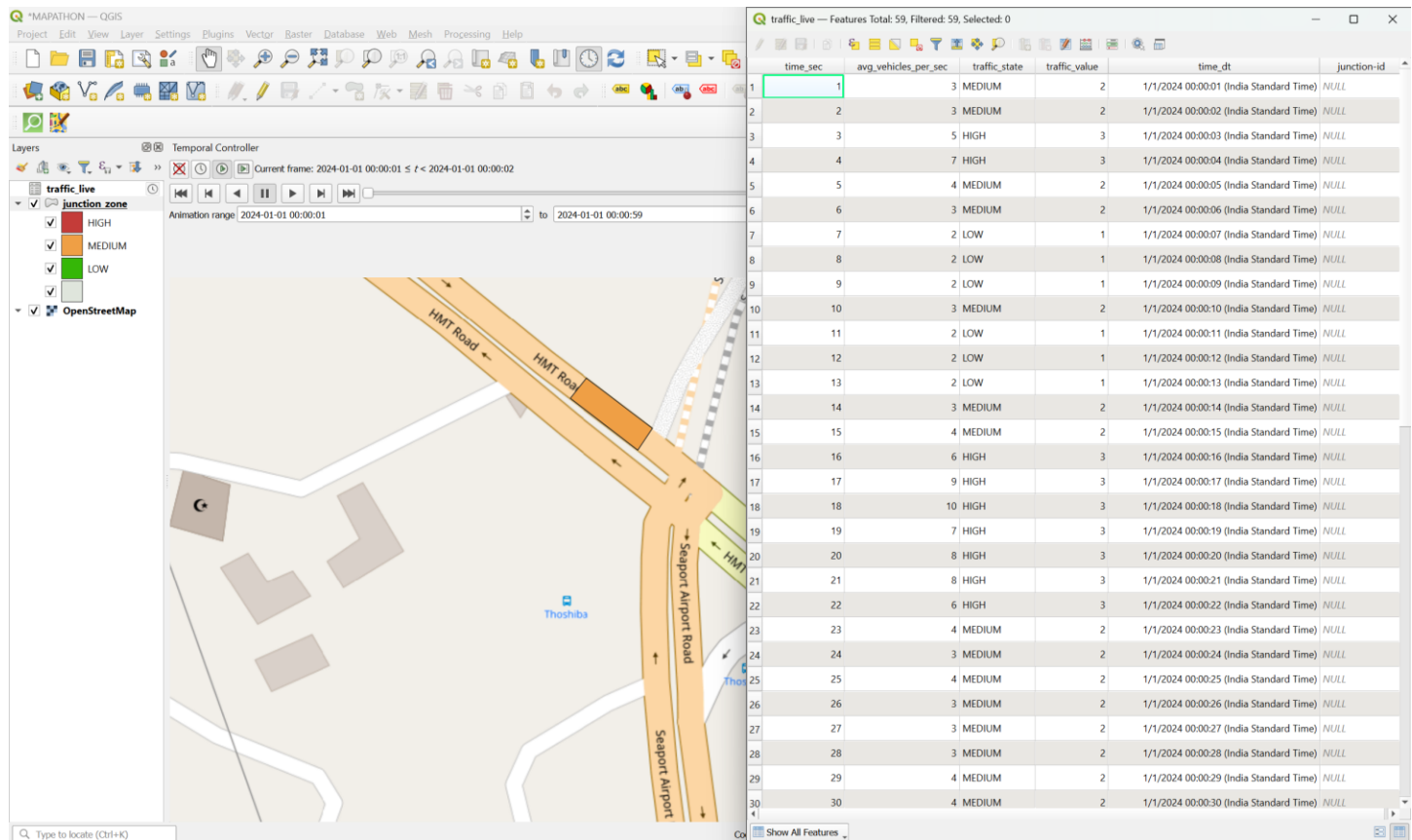
For each second of video, the following data is written to a CSV file:

- Time (seconds)
- Average vehicles per second
- Traffic state
- Traffic congestion value (1 / 2 / 3)
- Timestamp (for GIS temporal control)

| time_sec | avg_vehicles_per_sec | traffic_state | traffic_value |
|----------|----------------------|---------------|---------------|
| 1        | 3                    | MEDIUM        | 2             |
| 2        | 3                    | MEDIUM        | 2             |
| 3        | 5                    | HIGH          | 3             |
| 4        | 7                    | HIGH          | 3             |
| 5        | 4                    | MEDIUM        | 2             |
| 6        | 3                    | MEDIUM        | 2             |
| 7        | 2                    | LOW           | 1             |
| 8        | 2                    | LOW           | 1             |
| 9        | 2                    | LOW           | 1             |
| 10       | 3                    | MEDIUM        | 2             |
| 11       | 2                    | LOW           | 1             |
| 12       | 2                    | LOW           | 1             |
| 13       | 2                    | LOW           | 1             |
| 14       | 3                    | MEDIUM        | 2             |
| 15       | 4                    | MEDIUM        | 2             |
| 16       | 6                    | HIGH          | 3             |
| 17       | 9                    | HIGH          | 3             |
| 18       | 10                   | HIGH          | 3             |
| 19       | 7                    | HIGH          | 3             |
| 20       | 8                    | HIGH          | 3             |

## Step 6: Importing Data into QGIS

The CSV file is imported into QGIS as a temporal layer. The recorded traffic data is linked to the selected road segment or junction polygon created manually in QGIS.

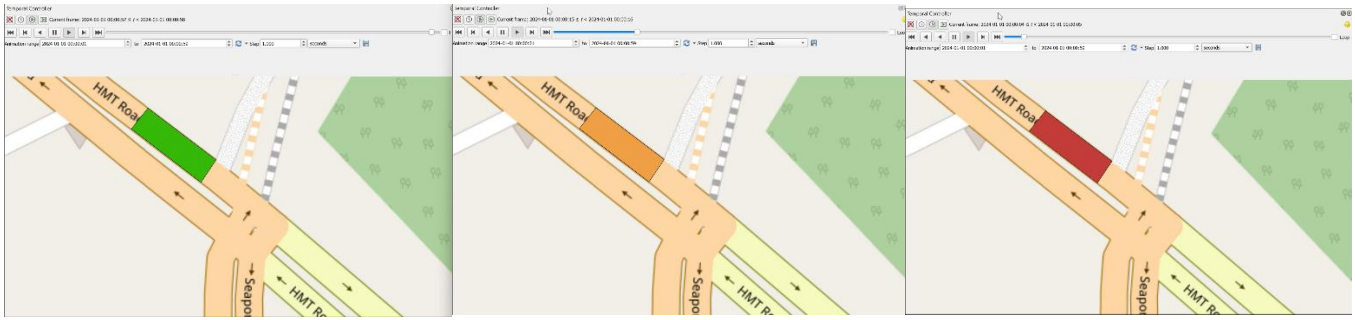


## Step 7: Spatial and Temporal Visualization

Using QGIS:

- A junction or road segment **polygon** is drawn manually
- Rule-based symbology is applied
- Traffic congestion values control polygon color:
  - Green → Low
  - Orange → Medium
  - Red → High
- QGIS Temporal Controller animates congestion changes over time

This enables visualization of how traffic conditions evolve every second.



## Documentation of Dashboard Functionalities and Analytical Outputs

### Dashboard Outputs

- Live vehicle counts per class
- Average vehicles per second
- Real-time congestion state
- Visual color-based congestion indicator

### GIS Outputs

- Selected road segments highlighted via polygons
- Color-coded congestion visualization
- Temporal playback of traffic changes
- Clear spatial representation of congestion hotspots

The combined dashboard and GIS outputs provide both operational and analytical insights.

### Policy and Architectural Recommendations

The system can be effectively deployed by traffic authorities for:

- Real-time traffic monitoring
- Junction-level congestion management
- Infrastructure planning and optimization

Future enhancements may include:

- Integration with live CCTV feeds
- City-wide multi-camera deployment
- Adaptive traffic signal control
- Cloud-based real-time data pipelines

The modular architecture allows scalability and integration into smart city traffic management systems.

### Conclusion

This project demonstrates the effective integration of deep learning, real-time visualization, and GIS for traffic congestion analysis. By using real-world video data and spatial visualization

techniques, GIS polygon mapping etc. The system provides actionable insights for both immediate traffic monitoring and long-term urban planning.

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